

SenseLess: Minimal Vision, Maximum Insight for Smart Homes

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Abstract—We present SenseLess, a hybrid anomaly detection framework for smart homes that, during the training phase, automatically labels images without manual annotation by combining sensor-guided detection, self-supervised visual clustering, and unsupervised multi-sensor delay estimation for precise alignment. During operation, the system relies primarily on non-vision sensors and activates a confidence-aware vision model only under low-confidence, thereby preserving privacy while maintaining adaptability. Evaluated in real home monitoring, SenseLess achieved an average label coverage of 97.65% with 94.9% accuracy and reduced vision usage to less than 4% of wall-clock operating time. Calibration mechanisms and minimal configuration requirements support scalability and deployment across diverse residential environments.

Index Terms—smart homes, anomaly detection, self-supervised learning, sensor fusion, image labeling, privacy preservation

I. INTRODUCTION

Anomaly detection in home care is vital for supporting older adults who live alone, where unnoticed events such as unattended appliances pose safety risks. Monitoring systems can enable early intervention but must operate continuously while preserving privacy and reducing caregiver burden. Effective solutions must adapt to changing behavior, limit false alarms, and provide actionable insights without intruding on daily life. Balancing these requirements is essential for safe and respectful in-home care.

Non-vision sensors are widely used in home care due to their low cost, unobtrusiveness, and continuous operation. Motion and door sensors detect abnormal activity and non-response [1]. Smart plugs, tags, and gas or smoke detectors enhance safety by detecting open doors, unattended stoves, or unauthorized access [2]. However, these systems often produce false alarms, lack context, and suffer from data drift [3]–[5].

Vision-based systems provide rich contextual information and have been applied to illness monitoring [6], fall detection using RGB and 3D video [7], and behavioral anomaly recognition [8]. However, they require large annotated datasets and are sensitive to privacy concerns, lighting changes, and limited camera coverage. Consequently, vision-based solutions remain difficult to adopt in personal living spaces, where privacy and user trust are critical.

Hybrid approaches combine multiple sensing modalities to improve robustness and coverage [9]. However, they often inherit the limitations of individual sensors and introduce additional computational cost, as well as challenges in data fusion and synchronization. Effective designs must therefore

integrate complementary signals while adapting to uncertainty, preserving privacy, and operating with minimal supervision.

To address these limitations, we propose a hybrid anomaly detection framework that combines non-vision data with self-supervised learning to build a vision model. During training, an unsupervised sensor-based model detects anomalies in environmental data and aligns them with camera frames to generate initial image labels. In parallel, a contrastive self-supervised method learns visual representations from unlabelled images and produces pseudo-labels through clustering. These outputs are fused using a confidence-weighted refinement process to create a high-quality self-labeled dataset for training a vision-based anomaly detector. During deployment, the non-vision model runs continuously, while the vision model is activated only when sensor predictions are uncertain. This design enables scalable anomaly detection with minimal visual exposure, no manual annotation, and ongoing adaptation to changing home environments, forming the basis of the proposed SenseLess framework.

In designing SenseLess, we focus on four challenges that arise when deploying anomaly detection in real-world homes:

- **Scarcity of annotated image data.** Vision-based anomaly detection relies on large labeled image sets, which are difficult to obtain in home environments. Manual annotation is costly, time-consuming, and often infeasible for rare or privacy-sensitive events, as it requires human involvement [10]. To address this, we adopt a dual-labeling strategy:
 - *Sensor-guided image labeling:* Non-vision sensors detect anomalies, which are aligned with camera frames to generate initial labels automatically.
 - *Self-supervised visual labeling:* A contrastive learning model learns representations from unlabelled images and clusters them to generate pseudo-labels based on visual similarity.
- **Temporal misalignment between sensor and image.** Sensors such as CO₂ respond with delay to environmental changes, which complicates alignment with visual observations [11]. Our analysis confirms these delays (Section II-B). We address this issue using backward synchronization that estimates and compensates for sensor response lag, improving labeling accuracy.
- **Balancing privacy with contextual awareness.** Always-on cameras are intrusive. Our system activates vision only

when sensor predictions are uncertain, limiting exposure while retaining interpretability.

- **Drift in sensor performance.** Sensor readings can shift over time with environmental or behavioral changes, reducing accuracy [5]. We address this with a self-healing mechanism where the vision model retrain the sensor model when drift is detected.

This work is guided by three research questions:

- **RQ1:** Can non-vision sensors be used to automatically generate accurate labels for training vision-based anomaly detection models?
- **RQ2:** How can self-supervised learning be combined with non-vision data to refine these labels and eliminate the need for manual annotation?
- **RQ3:** How can multi-modal sensor streams with variable response times be calibrated and aligned for accurate, synchronized anomaly detection?

To address these questions, we contribute the following:

- We propose *SenseLess*, a hybrid anomaly detection system that uses sensor-based and self-supervised pseudo-labels to train vision models without manual annotation.
- We introduce the Hierarchical Event-Driven Synchronization (HEDS) algorithm, which performs unsupervised multi-sensor delay estimation and compensates for response lags during sensor-to-image alignment. This improves label precision across heterogeneous modalities, ensuring reliable training data without manual calibration.
- We present a self-healing feedback mechanism in which the vision model retrain and recalibrates the non-vision model when drift or low-confidence conditions are detected, enabling long-term robustness.

The implementation of *SenseLess* is publicly available at <https://github.com/norahb/Senseless>.

II. SYSTEM DESIGN

A. Problem Definition

We consider the problem of automatically labeling images captured in indoor environments to train a vision-based anomaly detection model, without relying on manual annotation. Let $I = \{i_1, i_2, \dots, i_n\}$ be a set of unlabelled images, and $S = \{s_1, s_2, \dots, s_m\}$ be the corresponding aligned non-vision sensor readings associated with each image. The goal is to generate a refined label set $L^* = \{\ell_1, \ell_2, \dots, \ell_n\}$, where each label $\ell_k \in \{\text{Normal}, \text{Anomaly}, \text{Unknown}\}$, such that the resulting dataset $\{(i_k, \ell_k)\}$ can be used to train a robust, privacy-aware vision-based anomaly detector.

This task involves several key challenges:

- **Sensor-to-image alignment:** Sensor and image data must be temporally aligned despite variable response delays across sensor types.
- **Sensor label uncertainty:** Labels generated from unsupervised sensor models may be noisy or uncertain due to environmental drift or hardware variability.

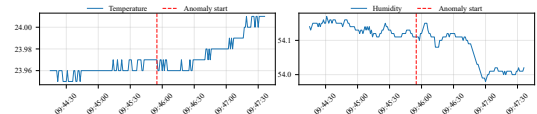


Fig. 1. Delayed temperature and humidity response after door opens. The red dashed line marks the event onset.

- **Unsupervised vision labeling:** Visual features extracted via self-supervised learning must be clustered and interpreted without access to ground truth.
- **Deployment constraints:** The final system must operate under privacy constraints, manage data drift, and provide confidence-aware decisions with minimal supervision.

B. Preliminary Investigation

Before system design, we ran experiments with real-world home care sensor data. They revealed two key challenges that shaped *SenseLess*: the limited accuracy of non-vision anomaly detection without labels, and sensor response delays that complicate alignment with visual input.

a) *Limitations of non-vision anomaly detection.*: We first trained an unsupervised anomaly detection model using only non-vision environmental data. Although effective in controlled settings, it performed poorly in real home environments. The lack of ground truth hindered validation of anomaly timing and cause, and many sensor patterns were ambiguous without context. These findings motivated adding an image stream and a self-supervised model to provide complementary cues for labeling.

b) *Sensor delay and misalignment.*: We analyzed sensor responses around events and observed that some environmental changes occurred gradually rather than immediately. For example, temperature and humidity often shifted several seconds after a door opened, as shown in Figure 1. This misalignment increases the risk of linking anomalies to incorrect image frames. Based on these observations, we designed the HEDS algorithm to detect and compensate for sensor delays during image alignment.

C. System Overview

SenseLess, illustrated in Figure 2, operates in two sequential phases: a training phase and a deployment phase.

During training, the system integrates three modules. First, a non-vision module is trained on normal sensor data to detect anomalies. Detected events are temporally aligned with images using the HEDS algorithm, which estimates and compensates for sensor delays. This aligns anomalies with their true visual context and improves labeling accuracy. Second, a self-supervised vision module extracts features from unlabelled images and generates pseudo-labels via clustering. Third, a label refinement module fuses sensor- and vision-derived labels using confidence weighting, producing a dataset for training a vision-based anomaly classifier.

In deployment, the non-vision module runs continuously as the primary detector. When predictions are uncertain, the vision module is selectively activated to validate the decision.

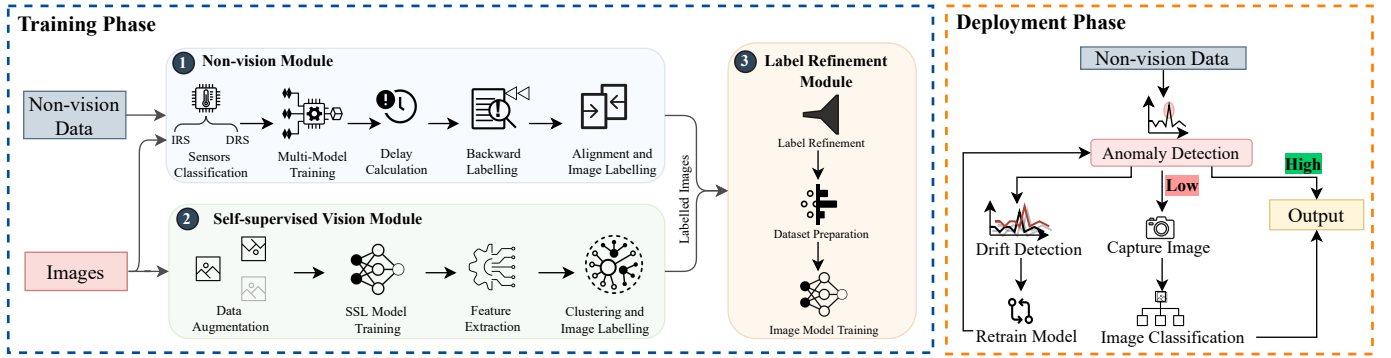


Fig. 2. System architecture of *SenseLess*. The training phase integrates non-vision anomaly detection, self-supervised vision, and label refinement to generate labeled data. The deployment phase relies primarily on non-vision sensing and selectively activates the vision module when confidence is low.

This selective activation preserves privacy by minimizing visual exposure and supports adaptation through feedback-driven retraining. Together, these modules enable *SenseLess* to deliver accurate and privacy-aware anomaly detection.

D. Training Phase

1) *Non-Vision Module*: The non-vision module generates initial labels for image data by detecting anomalies in environmental sensor streams. This process is implemented using the HEDS algorithm, which synchronizes sensor data with image timestamps while accounting for response delays and alignment uncertainty. The algorithm is outlined in Algorithm 1 and described in detail below.

a) *Sensor Classification*: Sensors are grouped into instant-response (IRS) and delayed-response (DRS) categories based on their characteristics and observed response behavior. IRS sensors (e.g., motion detectors) react immediately and are assigned zero delay ($\Delta t_s = 0$). DRS sensors (e.g., temperature or humidity) exhibit measurable latency and are assigned a calibrated delay $\Delta t_s > 0$ estimated during delay calculation.

b) *Model Training*: We train a multi-model ensemble of an autoencoder, Isolation Forest, and Elliptic Envelope, each capturing different aspects of anomaly behavior. Sensor data is cleaned in two stages: values outside physical ranges are replaced by interpolation, and statistical outliers are removed using a configurable method, with median absolute deviation as the default and Z-score filtering as an alternative. The autoencoder is trained on clean, normal-only data using mean squared error loss, with architecture scaled to sensor dimensionality. Features are normalized with either RobustScaler or StandardScaler, and thresholds are derived from the training error distribution and calibrated on validation data. During evaluation, anomaly scores from the three detectors are combined, and sensitivity is tuned by requiring agreement from one, two, or all detectors. This mechanism is optimized to maximize recall while limiting false positives. Confidence scores $c_k^{(s)}$ are computed from reconstruction errors and sensor-specific error distributions, aggregated across sensors, and calibrated with isotonic regression to produce reliable probabilities. When one detector achieves higher validation

performance than the others, the system can use it alone. If no method is clearly superior, the full ensemble is retained.

c) *Delay Estimation and Calibration*: For DRS, the system estimates the delay Δt_s between the anomaly onset and the point when reconstruction error first exceeds the threshold. Anomalies are detected with the trained model, which produces per-sensor error traces and thresholds. For each event, the anomaly start time t^{start} is identified by tracing backward from the error rise to where values stabilize near baseline, and the threshold crossing time t^{cross} is the first index where the error exceeds the threshold. The onsite delay is then computed as $\Delta t_s = t^{\text{cross}} - t^{\text{start}}$. The final Δt_s for each sensor is the mean of the filtered values, while IRS sensors are assigned $\Delta t_s = 0$. If ground truth event times are available, an optional lab-based calibration can be performed. In this case, delays are measured as the average difference between known event times and the first detected anomalies. Lab and onsite values may then be blended using configurable weights or adjusted relative to a reference sensor. Human-in-the-loop review is also supported, allowing experts to inspect and correct selected cases. This multi-stage calibration process ensures accurate temporal alignment between sensor events and images, improving label precision across all sensor types.

d) *Backward Labeling*: An anomaly period P_s is a contiguous segment where the reconstruction error of sensor s exceeds its learned threshold, with the onset given by the first sample of that segment. For each sensor s , the calibrated delay Δt_s shifts the start of P_s backward to obtain P'_s . All timestamps $t_k \in P'_s$ are labeled anomalous ($\ell_k^{(s)} = \text{Anomaly}$). The global label ℓ_k is set to Anomaly if any sensor flags an anomaly; otherwise, it is set to Normal.

e) *Image Alignment*: Image-sensor synchronization is performed using a dual-window alignment strategy. For each image captured at timestamp t_i , the system first searches for sensor anomalies within a *primary window* of $[t_i - 2, t_i + 2]$ seconds. This window accounts for timestamp jitter and corresponds to approximately two sensor sampling intervals at 1 Hz. If no anomaly is found, a *fallback window* is applied to handle delayed sensor responses. The fallback window size is defined as $d = \max(\Delta t_s, 5s)$, where Δt_s is the estimated delay for

Algorithm 1 Hierarchical Event-Driven Synchronization (HEDS)

```

1: Input: Sensor dataset  $S$ , image dataset  $I$ 
2: Output: Initial labeled image dataset  $L_s$ 
3: Step 1: Sensor classification
4: for all sensor  $s$  in sensor modalities do
5:   if  $s$  exhibits an immediate response to state changes then
6:     Assign  $s$  to the instant-response sensor (IRS) group and set  $\Delta t_s = 0$ 
7:   else
8:     Assign  $s$  to the delayed-response sensor (DRS) group
9:   end if
10: end for
11: Step 2: Model training
12: Train anomaly detection model  $f$  on a filtered normal-only subset of  $S$ 
13: Use  $f$  to compute sensor-based confidence scores  $c_k^{(s)}$  for detected anomalies
14: Step 3: Anomaly detection and delay estimation
15: for all sensor  $s$  in available sensors do
16:   Detect anomaly events for sensor  $s$  with start time  $t^{\text{start}}$ 
17:   if  $s$  is DRS then
18:     Find threshold crossing  $t_s^{\text{cross}}$  (first error > threshold for sensor  $s$ )
19:     Compute delay  $\Delta t_s = t_s^{\text{cross}} - t^{\text{start}}$ 
20:     Set  $\Delta t_s$  to mean of valid delays for sensor  $s$ 
21:   else
22:     Set  $\Delta t_s = 0$ 
23:   end if
24: end for
25: Step 4: Backward labeling and multi-sensor aggregation
26: for all sensor  $s$  in sensor modalities do
27:   for all anomaly periods  $P_s$  do
28:     Shift start time backward by  $\Delta t_s$  to obtain  $P'_s$ 
29:     for all  $t_k \in P'_s$  do
30:       Assign sensor-level label  $\ell_k^{(s)} = \text{Anomaly}$ 
31:     end for
32:   end for
33: end for
34: for all timestamps  $t_k$  do
35:   if  $\exists s : \ell_k^{(s)} = \text{Anomaly}$  then
36:     Set global label  $\ell_k = \text{Anomaly}$ 
37:   else
38:     Set  $\ell_k = \text{Normal}$ 
39:   end if
40: end for
41: Step 5: Sensor-to-image alignment
42: for all image timestamps  $t_i \in I$  do
43:   Define primary window:  $[t_i - 2, t_i + 2]$  seconds
44:   Search for anomaly event within primary window
45:   if match found then
46:     Assign  $\ell_k^{(s)}$  and  $c_k^{(s)}$ ; mark as Primary
47:   else
48:     Define fallback window:  $[t_i - d, t_i + d]$ , where  $d = \max(\Delta t_s, 5s)$ 
49:     Search for anomaly event within fallback window
50:     if match found then
51:       Assign  $\ell_k^{(s)}$  and  $c_k^{(s)}$ ; mark as Fallback
52:     else
53:       Assign  $\ell_k^{(s)} = \text{Unknown}$  with  $c_k^{(s)} = 0$ 
54:     end if
55:   end if
56: end for
57: return  $L_s$ 

```

sensor s , and 5 s corresponds to the maximum delay observed across all sensors in our dataset. Images matched within the primary window are labeled `Primary`, while those matched only within the fallback window are labeled `Fallback`. If no sensor event is found in either window, the image is labeled `Unknown` with zero confidence.

Matched images inherit the sensor-derived label $\ell_k^{(s)}$ and its confidence $c_k^{(s)}$. Unknown images are resolved during label refinement using SSL pseudo-labels or optionally escalated for human annotation. This process yields the labeled dataset $L_s = \{(i_k, \ell_k^{(s)}, c_k^{(s)})\}$, enabling supervised training without manual annotation.

2) *Self-Supervised Vision Module:* The self-supervised vision module learns visual representations from unlabelled images and produces cluster-based pseudo-labels independent

of the sensor subsystem.

We adopt a SimCLR framework with a ResNet-18 encoder and projection head. Each image is stochastically augmented twice (random crops, color jitter, grayscale, flipping) and the paired views are trained with the normalized temperature-scaled cross-entropy (NT-Xent) loss. This pulls representations of the same image together while separating others.

After training, the SimCLR encoder produces feature embeddings that are clustered using a Bayesian GMM with a Dirichlet prior. For each image embedding $\mathbf{x}_i$ assigned to cluster k , we compute confidence score using a hybrid formulation:

$$c_i^{(\text{ssl})} = \frac{1}{2} \left(P(k | \mathbf{x}_i) + \left(1 - \frac{d(\mathbf{x}_i, \boldsymbol{\mu}_k)}{d_{\text{max}}} \right) \right). \quad (1)$$

Here, $P(k | \mathbf{x}_i)$ is the GMM posterior probability, $d(\mathbf{x}_i, \boldsymbol{\mu}_k)$ is the Euclidean distance from sample to cluster mean, and d_{max} is the maximum distance observed in the dataset. This score yields confidence values in $[0, 1]$.

The result is a pseudo-labeled dataset (L_{ssl}) where each image has a cluster index and confidence score. These labels are later fused with sensor-derived labels during refinement.

3) *Label Refinement Module:* Each image i_k receives two candidate labels: a sensor label $\ell_k^{(s)}$ with confidence $c_k^{(s)}$ and an SSL label $\ell_k^{(\text{ssl})}$ with confidence $c_k^{(\text{ssl})}$. Confidence values are calibrated using isotonic regression models trained for each source and reused in later runs. Labels are then refined by the following rules:

- 1) If both confidences fall below thresholds ($\tau_s, \tau_{\text{ssl}}$), assign `Unknown`.
- 2) If both labels agree and at least one confidence exceeds its threshold, accept the shared label.
- 3) If only one source exceeds its threshold and is more confident, select its label.
- 4) Otherwise, assign `Unknown`.

Thresholds τ_s and τ_{ssl} are obtained through isotonic calibration against ground truth correctness flags, with the final cutoff selected by grid search to balance accuracy and coverage. Optional human review is supported for `Unknown` cases. Reviewed labels are stored to avoid duplication in future runs. The result is the final dataset $L^* = \{(i_k, \ell_k)\}$ with $\ell_k \in \{\text{Normal}, \text{Anomaly}, \text{Unknown}\}$.

4) *Vision Model Training:* We use the final labeled dataset L^* to train a supervised convolutional neural network for the detection of visual anomalies, excluding `Unknown` samples. We adopt MobileNetV2 for its favorable balance between classification performance and computational efficiency. The model is initialized with ImageNet weights, with the final layer adapted for binary classification. Training is performed using cross-entropy loss and the Adam optimizer, with early stopping based on validation loss. To improve generalization, we apply data augmentations such as random cropping, horizontal flipping, and color jitter.

E. Deployment Phase

During deployment, *SenseLess* runs a two-stage decision pipeline. The non-vision module operates continuously as the

primary anomaly detector, and the vision module is selectively activated when additional validation is required. This design minimizes visual exposure while maintaining robustness under uncertainty and sensor drift.

1) *Primary Non-Vision Module*: The non-vision module operates continuously, analyzing sensor streams and identifying outliers using an ensemble model trained on normal data. Sensor values are cleaned using a Z-score filter, and anomaly predictions are produced only for non-corrupted segments. Each prediction is assigned a label (Normal, Anomaly, or `Sensor_Error`) and accompanied by a confidence score estimated from reconstruction error. When the confidence is high, the prediction is accepted without further processing. Sensor errors and the names of affected sensors are logged per instance to support inspection.

2) *Fallback Vision Module*: When the non-vision module produces low confidence predictions or sensor errors, the system activates the vision module to perform secondary validation. In this case, a new image is captured and processed by a pre-trained classification model. The image is preprocessed and evaluated by the model, which returns a predicted class label (Normal or Anomaly) along with a confidence score derived from softmax outputs or object count estimation. Vision-based predictions that exceed a configured confidence threshold are logged and can override the original decision. This selective activation strategy improves robustness in ambiguous cases while minimizing unnecessary visual processing. The captured image is processed solely for fallback inference and is not retained, ensuring privacy is preserved.

3) *Data Drift Handling*: To support long-term reliability, the system maintains a reference baseline that characterizes normal sensor behavior under stable conditions. This baseline is computed during an initial data collection period or periodically updated using recent deployment logs filtered to exclude sensor errors. For each sensor, the system calculates mean, standard deviation, and percentiles, storing these as reference statistics. Once deployed, incoming sensor readings are compared against this baseline to detect data drift. Drift is flagged when recent samples exhibit significant deviation in statistical properties, such as shifts in mean or variance, or an increased outlier rate relative to the baseline. In parallel, the system monitors reconstruction error from the autoencoder to detect structural changes in sensor patterns that may not affect first-order statistics. When drift is detected consistently across multiple runs, the system activates a self-healing feedback loop. If the vision module is triggered due to low confidence and returns a high-confidence prediction that contradicts the non-vision decision, this prediction is logged and used as a trusted label. These vision-verified events are then used to retrain the non-vision model, ensuring that the system remains adaptive to changing environments without relying on manual annotations.

4) *Model Retraining*: When persistent data drift is detected, the system triggers a retraining workflow to update the non-vision anomaly detector. A data collector module first gathers a retraining set composed of recent sensor data, fallback vision-

labeled samples, replay buffer entries, and a small portion of the original training data. These sources are combined and filtered to ensure statistical quality and coverage diversity. A new non-vision model is then trained using the same ensemble configuration as in the original deployment, and its performance is evaluated against a validation set. If the new model outperforms the current one by a predefined margin (e.g., ≥ 0.02 macro-F1 improvement), it is deployed to replace the old model after creating a backup. To prevent overfitting to recent events, the replay buffer maintains representative samples across time and label types. This modular pipeline ensures that retraining occurs only when sufficient, high-quality data is available and performance gains are measurable.

III. IMPLEMENTATION AND EVALUATION

A. Evaluation Setup and Dataset Overview

We evaluated *SenseLess* in a real home equipped with environmental sensors and RGB and thermal cameras over four weeks of continuous operation. The evaluation covered four anomaly scenarios: open doors, appliance misuse, abnormal occupancy patterns, and obstructed pathways. Door and obstructed pathway anomalies were intentionally induced by leaving doors open for extended periods and placing objects to block paths. Appliance and occupancy anomalies were partly instructed, such as leaving the hob on unattended or remaining seated to raise CO₂ levels, and partly occurred during normal daily routines, capturing both controlled and realistic conditions.

The setup included temperature, humidity, pressure, CO₂, and ultrasonic distance sensors, with data collected at 1 Hz during 12.5 hours of daily operation. Ground truth labels were generated automatically using simple, case-specific mechanisms. For door, appliance, and occupancy scenarios, magnetic switches provided binary ground truth signals (open/closed, on/off, occupied/unoccupied). For obstructed pathways, labels were produced using a rule-based method based on ultrasonic distance measurements, where distances below a fixed threshold (e.g., 130 cm) indicated obstruction. These labels were used only for evaluation and calibration, not for training.

The dataset was split into 24 days for training and 6 days for testing. The image dataset contained 2,516 door images, 2,575 appliance images, 1,511 occupancy images, and 1,960 abnormal object images. In practice, deployments begin with a 1–2 week period of normal data collection to train the initial sensor model and establish baseline behavior.

B. Non-vision Sensor Model Performance

Table I reports the results on temporally held-out test sets. Non-vision models achieved high accuracy and macro-F1 on appliance and object use cases. Occupancy and door detection were less reliable. Across door, appliance, and occupancy cases, a single autoencoder consistently outperformed the ensemble variant, indicating that additional methods introduced more noise than benefit in these settings.

In the door case (Figure 3a), temperature, humidity, and pressure exhibited low variance and strong distributional

TABLE I
NON-VISION MODEL PERFORMANCE BY USE CASE

Use Case	Method	Accuracy	Macro-F1
Door	Single Autoencoder	86.0%	0.73
Appliance	Single Autoencoder	100%	0.98
Occupancy (Incremental)	Single Autoencoder	98.0%	0.95
Abnormal Object	Rule-based	100%	1.00

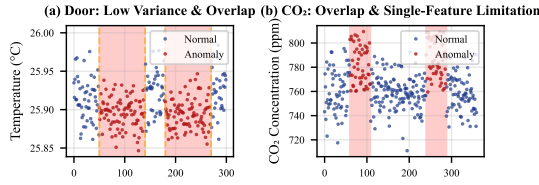


Fig. 3. Signal factors affecting detection. (a) Door anomalies show subtle, overlapping changes (b) CO₂ anomalies overlap strongly with normal values, highlighting the limitation of using a single feature. Red: anomaly periods.

overlap between normal and anomalous states. For example, anomaly means were close to normal averages. Correlations across features were weak, and temporal patterns followed smooth trends rather than abrupt shifts. These subtle changes reduced the contrast available to the autoencoder, making boundary precision and anomaly detection more difficult.

Occupancy detection using only CO₂ (Figure 3b) was hindered by strong overlap between normal and anomalous ranges. Anomalies averaged 791 ppm (SD = 67.7) and normal periods 725 ppm (SD = 42.4), making elevated levels common in both states. Additionally, this single-feature setup limited the autoencoder’s ability to distinguish anomalies. Temporal patterns showed smooth drifts rather than sharp jumps, further reducing discriminability. Conventional full-dataset training produced unstable boundaries, but incremental training in 5-day cycles with replay memory improved robustness, raising macro-F1 from 0.72 to 0.95.

C. Delay Calculation

Delay baselines were established using labeled datasets and a supervised Random Forest classifier (RF), which served as a proxy ground truth for anomaly onset. For each use case, the classifier was trained to detect the first anomaly, and the average gap between event onset and first detection defined the lab delay. These values provide a reference baseline but are not directly usable in deployment since true physical onset cannot be measured at scale. When evaluated against the RF baseline,

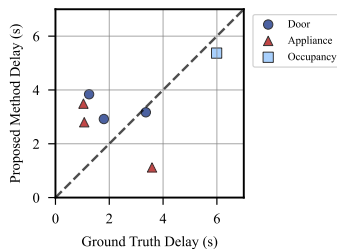


Fig. 4. Comparison between proxy ground truth (RF) delays and the proposed delay estimation method. Each point represents a sensor measurement: door (circles), appliance (triangles), and occupancy (square). The dashed line indicates close agreement ($y = x$).

our delay estimation achieved RMSE = 1.83s and MAE = 1.60s (Figure 4). Door events showed strong agreement, with

deviations below 1.2s for temperature and humidity, and 2.6s for pressure. Appliance sensors tracked short delays with errors between 1.7s and 2.5s, while the occupancy case differed by only 0.62s. These results confirm that onsite estimation provides physically plausible and deployment-ready delays, even in scenarios where environmental responses are gradual.

Door sensors showed close agreement between RF baselines (3.36s, 1.80s, 1.25s for temperature, humidity, and pressure) and onsite estimates (3.17s, 2.92s, 3.84s). For appliances, temperature and humidity delays were near-instantaneous (RF: 1.07s, 1.04s; onsite: 2.80s, 3.49s), while CO₂ exhibited slightly longer but consistent lags (RF: 3.59s, onsite: 1.12s). The occupancy case showed the slowest dynamics, with CO₂ delays of 5.99s (RF) and 5.37s (onsite), reflecting gradual diffusion compared to the sharper transitions of appliance events. If available, final delays can be blended with RF baselines, reference-sensor adjustments, or human-in-the-loop review to ensure robust alignment across deployment settings.

Across all use cases, onsite estimates stayed within seconds of RF-derived ground truth, while alternative unsupervised baselines such as CUSUM or cross-correlation often mis-estimated delays by minutes or produced negative lags. The stability of our method ensures physically plausible estimates, which is essential for sensor–image alignment in multimodal anomaly detection systems.

D. Sensor-Image Alignment

TABLE II
SENSOR-IMAGE ALIGNMENT RESULTS ACROSS USE CASES

Use Case	Match Rate (%)	Accuracy (%)
Door	100	67.2
Appliance	100	97.1
Occupancy	100	78.6
Abnormal Object	100	100

Alignment accuracy was evaluated by comparing the image labels transferred from non-vision records with the ground truth image labels (Table II). Since the alignment procedure relies on previously labeled sensor data, its performance reflects the effectiveness of the non-vision anomaly detector. Appliances achieved 97.1% accuracy, consistent with strong upstream sensor performance. Occupancy reached 78.6%, where the autoencoder struggled to capture CO₂ deviations, leading to reduced image-level agreement. Door alignment was lowest at 67.2%, mirroring the difficulty of detecting gradual environmental transitions in sensor data. Abnormal object events were straightforward to align, achieving 100% accuracy due to the clear signal from instant-response distance sensors. Overall, the observed performance variations reflect the inherent challenges of the upstream anomaly detection task, and therefore the alignment procedure performs best for events producing immediate and unambiguous sensor changes.

E. Self-Supervised Vision Module Performance

We evaluated multiple SSL models, including Barlow Twins, DINO, FastSiam, MoCo, and SMOG. SimCLR achieved the best balance of accuracy, training time, and model size, and was selected for our framework. For clustering,

we compared K-Means, GMM, DBSCAN, and Agglomerative methods. The GMM was adopted as the default, as it adapts the number of active components through a Dirichlet prior, provides well-calibrated confidence scores, and offered the most consistent trade-off between accuracy and efficiency across use cases. Figure 5 shows the performance of the selected SimCLR

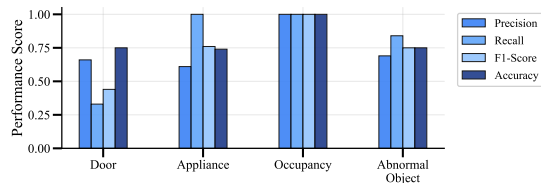


Fig. 5. Performance of the SimCLR and GMM pipeline.

+ GMM pipeline. Occupancy detection achieved perfect accuracy (1.00), with both classes classified correctly. Appliance detection reached 0.74 accuracy; anomaly recall was perfect (1.00) but precision was lower (0.61), indicating that some normal frames were misclassified as anomalies, likely due to thermal interference from nearby appliances. Abnormal object detection reached 0.75 accuracy with balanced F_1 -scores for both classes (anomaly 0.75, normal 0.71). Door detection was the most challenging, reaching 0.75 accuracy. Anomaly precision was high (0.66) but recall was lower (0.33).

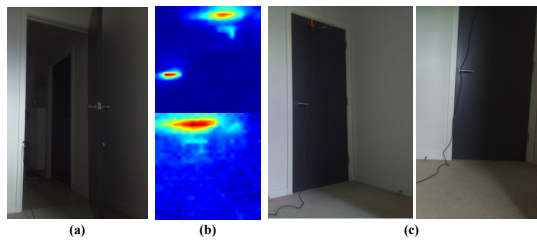


Fig. 6. Failure cases in SSL clustering: (a) overlapping door states, (b) thermal overlap from nearby appliances, (c) shadows misclassified as objects.

The observed errors were primarily due to visual ambiguity. In the door case, shadows and overlapping door states reduced class separability (Figure 6a). Appliance detection was affected by thermal interference from nearby devices (Figure 6b), while abnormal object detection was hindered by shadows misinterpreted as obstructions (Figure 6c). These failures illustrate how limited visual separability and intra-class variability constrain SSL clustering quality.

F. Label Refinement

Table III compares the non-vision, SSL, and hybrid methods. The non-vision approach achieved high coverage but variable accuracy, ranging from 67.2% in the door case to 100% for abnormal objects. SSL clustering offered more balanced performance, with stronger results for occupancy (100%) but lower accuracy in appliances and abnormal objects (74.0% and 75.2% respectively).

The hybrid method improved reliability by combining both sources. Overall, in automated mode, it achieved 97.65% coverage with 94.94% accuracy, yielding an effective accuracy of 92.50%. When low-confidence cases (Unknown) were reviewed, coverage reached 100% and accuracy rose to 95.84%,

with only 9.4% of door images requiring manual annotation. The appliance, occupancy, and abnormal object cases required no human input, as automated refinement already provided near-perfect performance.

Overall, the hybrid strategy reduced dependence on a single modality. Door events benefited from fusion, where SSL compensated for weak sensor signals, while appliance and abnormal object cases leveraged sensor reliability to correct SSL misclassifications. Occupancy detection benefited strongly from visual clustering, with SSL and the hybrid approach achieving perfect accuracy. These results demonstrate that multimodal refinement not only recovers accuracy lost in individual methods but also achieves robustness across diverse event types.

G. Image Classification Model Performance

We evaluated an image-only MobileNetV2 classifier on the door, appliance, and abnormal object use cases. For occupancy, we used a pre-trained EfficientNet-based crowd counting model without additional training. For each image classification task, we compare models trained using images automatically labeled by the proposed sensor-SSL pipeline with models trained using ground-truth image labels as a fully supervised image-only baseline. As shown in Fig. 7, training with automatically labeled data achieves strong performance, with accuracies of 91.67% (door), 95.74% (appliance), and 90.31% (abnormal object). Fully supervised training using ground-truth labels yields higher accuracy, reaching 93.65%, 100%, and 91.84%, respectively. This comparison shows that image-only models benefit from full supervision but require controlled data collection with manual labeling, whereas *SenseLess* attains competitive accuracy without manual annotation.

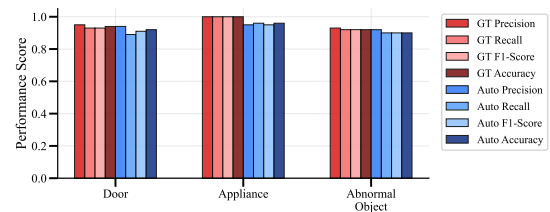


Fig. 7. Performance of image classification. GT denotes training with ground-truth labels, and Auto denotes training with automatically labeled data.

H. End-to-End Labeling Duration

Table IV reports the execution time of each stage in the hybrid labeling pipeline. Across the four use cases, anomaly detection was the most time-consuming step, ranging from 27.72 seconds (abnormal object) to 142.95 seconds (door). Image-sensor alignment required between 17.51 and 54.90 seconds, with larger datasets incurring longer alignment times. SSL labeling was comparatively fast, completing in 7.36–11.02 seconds across all scenarios. Label refinement varied depending on dataset complexity: while door required 73.99 seconds (including human labeling), appliance, occupancy, and abnormal object cases required less than one second each. Overall, full dataset annotation was achieved in 71.38–259.39

TABLE III
COMPREHENSIVE LABELING PERFORMANCE COMPARISON

Use Case	Non-Vision			SSL			Hybrid (Ours)					
							Automated			+ Human		
	Coverage (%)	Accuracy (%)	Effective Acc. (%)	Coverage (%)	Accuracy (%)	Effective Acc. (%)	Coverage (%)	Accuracy (%)	Effective Acc. (%)	Coverage (%)	Accuracy (%)	Effective Acc. (%)
Door	100	67.2	67.2	100	74.6	74.6	90.6	82.81	75.02	100	84.4	84.4
Appliance	100	97.1	97.1	100	74.0	74.0	100	96.97	96.97	100	96.97	96.97
Occupancy	100	78.6	78.6	100	100	100	100	100	100	100	100	100
Abnormal Object	100	100	100	100	75.2	75.2	100	100	100	100	100	100
Average	100.00	85.73	85.73	100.00	80.95	80.95	97.65	94.94	92.50	100.00	95.84	95.84

Notes: Coverage = Percentage of images labeled. Accuracy = Percentage correct among labeled images. Effective Acc. = Coverage $\times$ Accuracy. Automated = Fusion without human input. + Human = Including optional human labeling for ambiguous cases. **Bold** = Best performing method.

TABLE IV
LABELING PIPELINE TIMING PERFORMANCE (SECONDS)

Use Case	Non-Vision	Alignment	SSL	Refinement
Door	142.95	35.09	7.36	73.99*
Appliance	136.82	54.90	7.39	0.26
Occupancy	65.09	17.51	10.06	0.24
Abnormal Object	27.72	35.54	11.02	0.31
Average	93.15	35.76	8.96	18.70

*Includes human annotation for 9.4% (Door). Total pipeline time: 71.38–259.39 seconds.

seconds per use case, representing a significant efficiency improvement over manual labeling, which would require hours to days. These times exclude model training (non-vision anomaly detector and SimCLR).

I. Ablation on Delay Compensation

TABLE V
ALIGNMENT ACCURACY WITH AND WITHOUT HEDS DELAY COMPENSATION.

Use Case	HEDS Alignment (%)	No-Delay Alignment (%)
Door	67.2	62.5
Appliance	97.1	97.0
Occupancy	78.6	78.2
Abnormal Object	100.0	100.0

To quantify the contribution of the HEDS algorithm, we conducted an ablation study comparing alignment performance with and without delay compensation. Table V reports the accuracy of sensor-to-image label alignment across the four evaluation use cases.

HEDS improved label alignment in cases where sensor delays were present. The door scenario showed the largest gain, with accuracy increasing by 4.7%. Appliances benefited only marginally (+0.1%), reflecting their strong and immediate thermal signatures. Occupancy showed a small improvement (+0.4%). No improvement was observed for abnormal object detection, since instant-response distance sensors incurred negligible delays. These results demonstrate that backward delay compensation is most effective in scenarios affected by gradual environmental responses.

TABLE VI
VISION USAGE DURING DEPLOYMENT (6 DAYS, 12.5 H/DAY).

Use Case	Fallback Rate (%)	Inf. (ms)	Sec/day	Usage (%)
Door	27.2	39.5	483.7	1.08
Appliance	29.9	35.4	476.7	1.06
Occupancy	29.2	35.4	458.2	1.03
Abnormal Object	8.0	31.4	112.8	0.25
Overall	94.3	141.7	1531.4	3.42

J. Deployment Evaluation

We evaluated *SenseLess* over the final six days of deployment, covering 270k records per use case. Figure 8 and Table VI summarise accuracy, fallback behaviour, and vision usage. Appliance monitoring maintained strong performance (88.5%) with modest fallback use, while occupancy detection started at 21.1% accuracy but improved to 76.7% after re-training, though still requiring frequent fallback (29.2%). Door detection initially achieved 55.2% accuracy but recovered to 69.3% following automatic retraining, and abnormal object detection achieved perfect accuracy with minimal fallback.

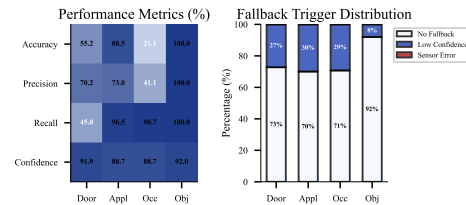


Fig. 8. Deployment performance: (left) detection metrics and confidence scores; (right) vision fallback triggers showing the share of decisions needing secondary validation due to low sensor confidence or errors.

Across all scenarios, fallback activation scaled with sensor uncertainty and drift, providing secondary validation only when required. Door and occupancy triggered the highest fallback rates, but each inference added just tens of milliseconds. Normalized over 270,000s of deployment (6 days, 12.5 h/day), the cumulative vision overhead was 9188s, amounting to 3.42% of operating time.

K. Cross-Home Validation

We further evaluated the framework in a new home using the door and occupancy use cases. For the door case, training on two days of data and testing on a third yielded 96.0% accuracy

with an F1-score of 0.84. During deployment, the initial model achieved 24.4% accuracy, but automatic retraining adapted within the same environment and improved performance to 80.8% without manual intervention. For the occupancy case, training on 2.5 days and testing on 1.5 days achieved 93.0% accuracy with an F1-score of 0.91, and deployment maintained 89.6% accuracy with anomaly recall of 99.2%.

IV. DISCUSSION

SenseLess shows that a hybrid sensor-vision pipeline enables privacy-aware, and adaptive anomaly detection in homes without manual image labeling. The HEDS algorithm corrects sensor delays and improves label accuracy for gradual responses. The hybrid labeling strategy achieves high coverage (97.65%) and accuracy (94.94%) with reduced annotation effort. Selective fallback limits vision use to less than 4% of operating time, supporting privacy.

The self-healing enables adaptation under data drift by retraining the non-vision model using vision-based corrections. Door and occupancy accuracy improve after retraining, rising from 55.2% to 69.3% and from 21.1% to 76.7%, respectively. Cross-home results show both generalization and its limits. Physics-driven signals such as CO₂ transfer well, while layout-dependent signals require adaptation. Overall, generalization is scenario-dependent but achievable through adaptation.

Still, our method has limitations that are reflected in the observed performance variations. Differences in non-vision sensor signals and availability lead to uneven sensitivity across scenarios. Strong and immediate signatures from appliances and distance sensors support accurate detection, while gradual transitions in door and occupancy cases reduce sensitivity. Sensor types, placement, sampling rates, and reliability vary across homes. Although the framework supports different sensor subsets, detection accuracy depends on signal quality and stability. Moreover, SenseLess focuses on anomalies manifested through environmental changes and does not directly address human-centered events, such as falls, which rely on different sensing dynamics. Supporting such events would require additional sensing modalities or explicit behavioral modeling. Environmental variability further affects performance, as differences in layout, ventilation, and occupant routines shift baseline behavior. Abrupt changes such as renovations or sensor relocation may temporarily reduce accuracy until adaptation occurs. The delay estimation algorithm assumes a clear anomaly onset followed by a delayed sensor response at 1-second sampling resolution. Weak or intermittent anomalies may violate this assumption. Lower sampling rates, such as minute-level data, reduce temporal resolution and would require architectural changes. In addition, although the vision component uses selective activation and multimodal validation to limit visual uncertainty, constraints remain. Training-time augmentations improve robustness to lighting changes and partial visibility through random resized cropping and color jittering. However camera coverage is limited, anomalies may be fully occluded, and reliance on single images restricts

interpretation of temporally evolving events, motivating multi-camera extensions. At the same time, expanding visual coverage directly raises privacy concerns. Prior work reports low acceptance of always-on cameras and higher acceptance when vision is restricted to specific situations and combined with privacy-preserving measures [12]. SenseLess adopts this trade-off by activating vision only under non-vision uncertainty. Although user preferences are not directly evaluated, the design aligns with reported privacy expectations.

V. RELATED WORK

A. Sensor-Based Anomaly Detection

Unsupervised and semi-supervised methods are common in smart home anomaly detection, including autoencoders [13] and Isolation Forest [14], with ensembles improving robustness [15]. Federated learning limits data sharing [16], but most approaches assume homogeneous sensors and ignore camera fusion and sensor delay [15].

Vision-based systems provide spatial context for fall and indoor anomaly detection [3], [17], [18], but require labeled data and raise privacy and computational concerns [19]. Surveys highlight open challenges in multimodal fusion, drift adaptation, and privacy-aware design [10]. No prior work combines ambient sensing and vision in a delay-aware, privacy-preserving, bidirectional framework.

B. Image Labeling in Vision-Based Systems

Labeling large-scale visual data is a major bottleneck, especially for anomaly detection. Manual annotation is accurate but costly and time-consuming [20], while semi-automated methods combine model suggestions with human input [21], [22]. Automated approaches rely on clustering [23], pre-trained models [24], or domain-specific pipelines [25].

Most methods assume well-defined categories, whereas anomalies are rare and context-dependent. Cross-modal labeling using wearable sensors has been explored for activity recognition [26]–[28], but remains limited for image-based anomaly detection [29], [30]. We extend this line by using non-vision signals to label indoor images, enabling training without manual supervision.

VI. CONCLUSIONS

We presented SenseLess, a hybrid anomaly detection framework that generates image labels without manual annotation by combining sensor-guided and self-supervised learning with delay-aware alignment. The system integrates confidence-aware vision fallback, drift detection, and selective vision activation to preserve privacy while maintaining adaptability across scenarios. Evaluation across four home monitoring tasks demonstrated high labeling coverage and competitive accuracy, with scalability to new environments supported by modular design and minimal configuration requirements.

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